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28 July 1993
SUPERSEDING
ATPD 2110A
30 August 1980

MILITARY SPECIFICATION

CARRIER, ARMORED, FULL-TRACKED: M113A3

This specification is for use within the U. S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the interface, performance and inspection for a full-tracked, diesel powered vehicle designed for cross country and amphibious operations.

1.2 Classification. Vehicles shall be of the following model, as specified:

Model	Army Drawing	Vehicle Type
M113A3	8750170	Carrier, Personnel

Beneficial comments (recommendations, additions, deletions, and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, Michigan 48397-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2350

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2. APPLICABLE DOCUMENT

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800	-	Fuel Oil, Diesel.

MILITARY

MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-C-45360	-	Carriers, Personnel, Full-Tracked: Armored, M113, M113A1, M113A2, M113A2/EFT, and M113A3; Mortars, Self-Propelled: 107 mm, M106, M106A1, and M106A2, and 81mm, M125A1 and M125A2; and Smoke Generator: M1059; Processing for Storage and Shipment of.
MIL-H-62315		Heater Assembly, Combustion Vehicular Compartment, 60,000 BTU/HR.
MIL-E-62541	-	Engine, Diesel, 6 Cylinder V-Type, 275 H.P, Turbocharged.

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STANDARDS

MILITARY

MIL-STD-130	-	Identification Marking for U.S. Military Property.
MIL-STD-810	-	Climatic Extremes for Military Equipment.
MIL-STD-461A, Notice 4	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462, Notice 3	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-642	-	Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1275	-	Characteristics of 28 C Volt DC Electrical Systems in Military Vehicles.
MIL-STD-1472	-	Human Engineering Design Criteria.
MIL-STD-1474	-	Noise Limits for Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

8750170	-	Carrier, Personnel, Full-Tracked, Armored, M113A3.
10949802	-	Fixed Fire Extinguisher.

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12350010	-	Chart, Lubricant, Fuel, Coolant, M113A3.
12259266	-	STE/ICE Test Set.

PURCHASE DESCRIPTIONS

ATPD-2109	-	Transmission, Hydraulic: Automatic X200-4.

(Copies of specifications, standards, and drawings required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article (Preproduction). The contractor shall furnish a sample unit for first article inspection and approval (4.4). First article sample shall be inspected by the contractor under the surveillance of the Government to determine conformance to the quality assurance provisions of this specification. The first article sample shall be fully representative of units to be supplied from production tooling and facilities.

3.2 Initial production vehicle. Vehicles manufactured at initiation of quantity production shall be inspected, and shall pass initial production inspection(4.2 and 4.4.2). Initial production vehicles shall be representative of vehicles to be furnished under this specification and shall include all current modifications. No modification shall be applied to subsequent vehicles without prior approval of the Government.

3.3 5,000 mile operational profile. The vehicle shall meet the operational requirements specified in 3.3.1 through 3.3.3 when operated for 5,000 miles, combat loaded (27,200 ± 200 pounds (LB) gross vehicle weight (GVW)), and operated at speeds appropriate to the terrain as follows:

a. 20% hard-surfaced roads
b. 40% secondary roads
c. 40% cross-country

(See 4.7.2)

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NOTE: During operation on hard-surfaced roads and cross-country, the vehicle shall be capable of towing a vehicle weighing 27,200 ± 200 LB GVW for 20 miles in each mode.

3.3.1 Reliability. Reliability shall be not less than 850 mean miles between failure (MMBF) during the first 5,000 miles, when operated in accordance with the operational profile (see 3.3). For calculating MMBF, the definition in the M113A2 family 50 mile mission failure scoring criteria shall be used (see 4.7.2.1 and 6.9).

3.3.2 Durability. The vehicle, with rated payload (GVW 27,200 ± 200 lb), shall complete the 5,000 test miles without replacement, rebuild, or major overhaul of the engine, transmission, or final drives, and without repair or rebuild of the hull, when operated in accordance with the operational profile (see 3.3 and 4.7.2.2).

3.3.3 Maintainability. Total maintenance, excluding driver/crew checks and services, shall not exceed 175 man-hours during 5,000 miles of operation in accordance with the operational profile (see 3.3). This equates to a Maintenance Ratio (MR) of 0.35 at 10 operational miles, equivalent to one operational hour. The scheduled maintenance interval shall be three months or 750 miles, whichever comes first (see 4.7.2.3).

3.4 Materials. The materials shall be as specified herein and in referenced documents. The materials shall be uniform in quality and free from defects and imperfections that might affect the serviceability and reliability of the finished product (see 4.1.1).

3.5 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPLs) wherever specified (see 4.1.2).

3.6 Construction. The vehicle shall be constructed in accordance with applicable drawings, specifications and standards, including mounting provisions for external armor kit. Completed vehicles shall meet all requirements specified herein, and be of a quality that assures a vehicle free of defects that would compromise, limit, or reduce its performance (see 4.1.3).

3.7 Physical characteristics.

3.7.1 Bearing seals. When fording, operating amphibiously or in mud, sand or snow, seals shall prevent the entrance of foreign matter into the bearings which are exposed to contamination during these operations. Bearing seals shall restrict the leaking of lubricants from bearings. Lubricant leakage at static and dynamic seals shall not be in excess of a Class I (see 4.7.6).

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3.7.2 Doors. All doors shall fit and function without interference or misalignment (see 4.7.7).

3.7.3 Hatches and vision devices. All hatches shall fit and function without interference or misalignment and have positive locks for the opened and closed positions. Vision device receptacles incorporated in the vehicle shall accept Government supplied vision devices without binding or interference and provide adequate support during all vehicle operating conditions. All closed hatches and installed vision devices shall individually restrict entrance of water to a maximum of 10 drops per minute into the vehicle (see 4.7.8).

3.7.4 Engine. The engine shall conform to MIL-E-62541 and when installed in the vehicle, shall operate throughout all speed ranges without evidence of coolant or fuel leakage. Lubricant leakage at static and dynamic seals shall not be in excess of a Class I (see 4.7.9 and 6.6.1).

3.7.5 Engine exhaust system. Engine exhaust leakage is permitted at connections provided clamping devices remain secure during vehicle operation (see 4.7.10).

3.7.6 Fuel system. The fuel tanks, lines and connections shall be internally clean prior to initial fueling of the vehicle. The fuel system shall be free of leakage. (see 4.7.11).

3.7.6.1 Fuel system, grades and slopes. During vehicle operation, the fuel system shall maintain a continuous supply of fuel to the engine when ascending and descending 60% grades in forward and reverse gear, and when the vehicle is being operated on 30% side slopes with either side of the vehicle up-slope (see 4.7.11.1).

3.7.6.2 Fuel tanks. The fuel tanks shall be capable of receiving fuel at the average rate of 25 gallons per 30 seconds and shall be free of leakage. (see 4.7.11.2).

3.7.7 Ramp hydraulic system. The ramp hydraulic system shall be internally clean prior to initial servicing of the vehicle. All hydraulic lines and connections shall be free of fluid leakage (see 4.7.12).

3.7.7.1 Ramp. The time required to raise the ramp from the fully lowered position to closed position, shall not be greater than 18 seconds with the engine operating at no more than 1200 rpm. The ramp locking mechanism shall be adjusted to permit single hand locking and unlocking operation (see 4.7.12.1).

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3.7.8 Transmission. The transmission shall conform to requirements of ATPD-2109. When installed in the vehicle, the transmission shall operate throughout all gear ranges without evidence of leakage in excess of a Class I at static and dynamic seals (see 4.7.13).

3.7.9 Electrical system.

3.7.9.1 Generating system. The generating system, with engine operating at 2,150 ± 100 rpm, shall provide 28.0 ± 0.5 volts (without load) and 27.3 ± 0.3 volts (with 180 ± 5 ampere load) at the vehicle batteries (se 4.7.14).

3.7.9.2 Lights. Internal and external lights shall operate, and all electrical contacts and connections shall maintain positive contact, under all vehicle operating conditions (see 4.7.15).

3.7.10 Heater. Personnel heater conforming to MIL-H-62315, when installed, shall start, operate and purge satisfactorily with the vehicle electrical system and diesel fuel conforming to VV-F-800 (see 4.7.16).

3.7.10.1 Heater exhaust system. The heater exhaust system, when installed, shall be free of leakage (see 4.7.16.1).

3.7.11 Communications and electrical circuits. Communication and electrical circuits shall maintain continuity from end to end, without shorts from pin to pin or from pin to connector shell. (see 4.7.17)

3.7.12 System voltage characteristics. Electrical system voltage characteristics shall fall within the limits for transient and steady state voltage requirements specified in MIL-STD-1275 (see 4.7.18).

3.7.13 Electromagnetic interference suppression. The vehicle, when operating, shall conform to the requirements for EMI/EMC suppression in accordance with emission requirements RE05 and CE07 of MIL-STD-461A, Notice 4. The vehicle horn and switching of headlights are excluded (see 4.7.19).

3.7.14 Simplified test equipment/internal combustion engine (STE/ICE) compatibility. For fault detection, the vehicle shall be compatible with STE/ICE test set 19207-12259266 (see 4.7.20).

3.7.15 Controls. All electrical and mechanical controls shall operate without malfunction throughout all ranges of operation under all operating conditions. (see 4.7.21)

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3.7.16 Adjustment mechanisms. All adjustment mechanisms shall be operable and provide adjustment capability within specified ranges (see 4.7.22).

3.7.17 Trim vane. The trim vane shall be operable in both the extended and retracted positions. (see 4.7.23).

3.7.18 Stowed equipment. All interior and exterior stowage items when stowed (combat loaded) on a completely assembled vehicle in the spaces provided, shall fit properly and not interfere with operation of the vehicle and components in any manner (see 4.7.24).

3.7.19 Spall liners. The spall liners shall conform to requirements specified on applicable drawings (see 4.7.25).

3.8 Performance. A complete vehicle, loaded with rated payload, serviced as specified in 3.8.1, and equipped for existing climatic conditions, shall operate as specified herein without special equipment. Performance shall be demonstrated on smooth, level, hard-surfaced roads free of loose material except as otherwise specified herein (see 4.7.4 and 4.7.5).

3.8.1 Preparation. Except for equipment and component installation and servicing performed by the contractor as a final processing of vehicles for acceptance, the vehicle shall be completely assembled and all adjustments completed. The vehicle shall be serviced with coolant, lubricant and fuel as specified on drawing 12350010. Specific applications by grades or types for various expected seasonal temperature ranges shall be as specified.

3.8.2 Oil pressure and temperature. Prior to vehicle operation, the engine shall be operated at 600 to 800 rpm until lubricating oil is at operating pressure and temperature. If at any time during vehicle operation, the engine has been stopped for at least 15 minutes, the engine shall again be operated, as above, before continuing vehicle operation.

3.8.3 Break-in run. (see 4.7.1)

3.8.3.1 Break-in run (track). Prior to tests, each vehicle shall be given a break-in run as specified in table I on smooth, level, hard-surfaced roads or a test track, except a section of the road shall be as shown in figure 1. After each division of break-in run, the vehicle shall be stopped and, after allowing the engine to idle for not less than one minute, driven in reverse for a distance of not less than 50 feet (see 4.7.1.1).

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TABLE I. Break-in run (track)

Division of run	Speed (mph)	Distance (miles)
a	0 to 10	5
b	11 to 20	5
c	21 to 30	10
d	31 to maximum governed	10

3.8.3.2 Break-in run (simulated). When authorized, a partially simulated break-in run may be given in lieu of the run specified in 3.7.3.1, using the Army Skid Type Dynamometer as specified in table II. After completion of each division "a" through "d", the vehicle shall be operated in reverse for 15 seconds minimum with engine rpm at 1,000 minimum (see 4.7.1.2).

TABLE II. Simulated break-in run.

Division of run	Speed (mph)	Distance (miles)	Run Conditions
a	0 to 10	5	Dynamometer
b	11 to 20	5	Dynamometer
c	21 to 30	5	Dynamometer
d	31 to maximum governed	5	Dynamometer
e	21 to 30	5	Track (road)
f	31 to maximum governed	5	Track (road)

3.8.3.3 Condition after run. After completion of the break-in run, there shall be no indication of damaged components, maladjustment, or loss of original adjustment that may cause faulty vehicle operation subsequent to Government acceptance.

3.8.4 Cooling systems.

3.8.4.1 Engine cooling. With the vehicle operating in an ambient temperature of 120°F measured at the inlet grille, and a tractive effort (TE) to gross vehicle weight ratio of .60 (16, 320 LB TE/27,200 LB GVW) measured at the sprockets, the engine oil temperature not greater than 275 degrees F, the engine cooling system using water as coolant shall maintain the engine coolant

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at a temperature not greater than 230°F, measured at a point between the coolant outlet from the engine and the transmission oil cooler inlet. The radiator pressure cap shall relieve the cooling system operating pressure between 12 and 16 pounds per square inch gage (psig) (see 4.7.26)

3.8.4.1.1 Deaeration and clearing capacity. The cooling system using water as coolant shall perform as specified while ingesting air at rates up to 0.6 cfm, for not less than 12 minutes. The system shall purge itself of air within 1 hour after stopping air injection (see 4.7.26.1).

3.8.4.1.2 Draw down capacity. The cooling system shall perform as specified while operating with a coolant volume loss of up to 6 quarts. (see 4.7.26.2)

3.8.4.2 Transmission lubricant cooling. With the vehicle operating in an ambient temperature of 120°F measured at the inlet grille and a tractive effort ratio of 0.60 (16,320 LB TE/27,200 LB GVW) measured at the sprockets, the transmission lubricant cooling system shall maintain lubricant temperature at not greater than 305°F, measured from the transmission outlet into the cooler (see 4.7.27).

NOTE: Requirements cited in paragraph 3.8.4 are not applicable to conversion programs unless otherwise specified by contract.

3.8.5 Bilge pumps. The bilge pumps shall remove water from the vehicle at a rate of not less than 44 gpm each with the pump motor operating on 28.0 volts maximum (see 4.7.28).

3.8.6 Speed and drift.

3.8.6.1 Speed. The vehicle shall sustain a speed of 40 mph on level ground after break-in, without damage to the power plant or suspension system (see 4.7.29).

3.8.6.2 Drift. With the vehicle traveling 25 mph ± 5 mph, drift shall not exceed three feet in 100 feet (see 4.7.30).

3.8.7 Acceleration. With the vehicle at a standstill position, the transmission in 1-4 range and the engine at idle speed, the vehicle shall accelerate from 0 to 20 mph in not more than 10.5 seconds (see 4.7.31).

3.8.8 Engine starting on slopes. The engine shall restart in not more than one minute when the vehicle is headed up or down a 60% slope, after sitting for a minimum of 1 minute. Similarly, the engine shall start when on a 30% side slope (see 4.7.32).

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3.8.9 Braking.

3.8.9.1 Stopping and vehicle control. The vehicle shall stop within a distance of 40 feet from the point of service brake application when traveling at a speed of 20 mph ± 1 mph, on dry, smooth, level, hard-surfaced roads free of loose material. The drift shall not exceed three feet in the stopping distance. Steering action is permitted during braking (see 4.7.33).

3.8.9.2 Holding. The vehicle shall hold stationary when parked, headed up or headed down, on a 60% longitudinal slope with the brakes locked for parking and with the transmission in neutral and the engine off (see 4.7.34).

3.8.10 Turning. With the transmission control in first range, the vehicle shall make continuous, full 360° turns to the right and to the left. At all times during the turns, the rear point of the outer edge of the vehicle shall be in contact with or within a 45 foot diameter circle (see 4.7.35).

3.8.10.1 Pivot steer. With the transmission control in pivot steer, the vehicle shall be capable of executing a 360° pivot turn within a 25 foot diameter circle in either direction (see 4.7.36).

3.8.11 Water crossing.

3.8.11.1 Leakage. During water operation, leakage of water into the hull of the vehicle shall be not more than one quart per minute (see 4.7.37).

3.8.11.2 Water operation. The vehicle, launched amphibiously, shall negotiate water of a depth sufficient to float the vehicle and with the height of waves not exceeding one foot. One foot height of wave is the height measured from trough to crest. When afloat in calm water, the vehicle shall attain a maximum speed of not less than 3 mph. The vehicle shall negotiate a 180 degree turn within a 30 foot corridor when crossing calm, open water (see 4.7.38).

3.8.12 Trench crossing. The vehicle shall cross in a forward direction, trenches measuring up to 36 inches in depth and up to 66 inches in width without functional damage to the vehicle (see 4.7.39).

3.8.13 Vertical obstacles. The vehicle, operating in forward gear, shall cross vertical obstacles (walls) measuring up to 24 inches in height and, in reverse gear, obstacles up to 14 inches in height without functional damage to the vehicle (see 4.7.40).

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3.8.14 Grade and slope operation. The vehicle shall ascend 60% dry grades in forward and reverse gear from a standstill position at the base of the grade without stalling or damage to the power plant or power train. The vehicle shall operate without malfunction or damage on 30% dry side slopes with either side of vehicle upslope. Proper oil pressure and temperature shall be maintained in the engine and transmission during all grade and slope operations. Oil pressure readings shall be made with the engine operating at 650 to 800 rpm (see 4.7.41).

3.8.15 Suspension. The vehicle suspension shall operate over not less than three 6 inch wide by 6 inch high obstacles, spaced 30 feet apart on a flat surface, at a speed of 18 to 22 mph, without the track drive sprockets or idler wheels impacting the hard surface (see 4.7.42).

3.8.16 Cupola. The installed cupola, without gun and ammunition, shall rotate (after breakaway) without binding under an applied torque of not more than 150 inch pounds with the vehicle level and the commander's hatch closed. The cupola travel lock shall hold the cupola in a locked position during all vehicle operating conditions (see 4.7.43).

3.9 Environmental conditions. The complete vehicle shall operate in ambient air temperatures of +120°F to -25°F without winterization equipment and with winterization equipment installed, down to -60°F. The complete vehicle, when in storage, shall withstand climatic extremes as specified in MIL-STD-810 without deterioration that may cause failure of any component part of the vehicle (see 4.7.44).

3.10 Noise level. Vehicle noise shall be in accordance with MIL-STD-1474, for all occupied areas of the vehicle (see 4.7.45).

3.11 Toxic fumes concentration. Concentrations of toxic fumes shall be in accordance with MIL-STD-1472, for all occupied areas of the vehicle (see 4.7.46).

3.12 Fire extinguisher. The fixed fire extinguisher system shall be readily accessible for servicing and for internal and external operation. Maximum effort for externally discharging system shall not exceed value specified on applicable drawing (10949802). (see 4.7.47 and 4.7.47.1).

3.13 Protective finish. Unless otherwise specified, protective finish for vehicle and component assemblies and parts shall be in accordance with applicable drawings (see 4.7.48).

3.14 Marking and data plates. The vehicle marking shall be in accordance with MIL-STD-642. Parts and components shall be marked in accordance with MIL-STD-130. Name, shipping and service data plates shall meet the requirements of MIL-P-514 or applicable drawings (see 4.7.49).

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3.15 Welding repairs. Welding repairs of any type or class shall be made only when, and to the extent, specifically authorized by the applicable drawings, specifications or welding procedures which have been approved by the procuring activity.

3.16 Workmanship. Workmanship shall be of such quality as to assure that the vehicle and its components are free of defects that compromise, limit, or reduce capability of its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities or inspection equipment suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Materials. Materials inspection and certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government. (see 3.7)

4.1.2 Qualified products. Contractor's inspection records shall specifically list all QPL items by number and date of the QPL, name of contractor and part or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document and name of contractor. (see 3.5)

4.1.3 Construction. Parts, components and assemblies shall be inspected in accordance with applicable drawings and specifications. (see 3.6)

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Initial production inspection (4.4.2)

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c. Control test (see 4.4.3)
d. Quality conformance inspection (see 4.5)
e. Comparison production test (see 4.6)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be conducted under the conditions of temperature, barometric pressure, and humidity prevailing at the site of test or place of manufacture.

4.4 First article inspection(Preproduction). The first article inspection shall be performed on sample initial production vehicle(s) (see 3.1) which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Preliminary Inspection and Acceptance. Preliminary Inspection and Acceptance shall include the inspections specified in 4.4.1.1 and 4.4.1.2.

4.4.1.1 Sample Size. A minimum of one vehicle shall be subjected to Preliminary Inspection and Acceptance inspection.

4.4.1.2 Government in-process examination. An in-process examination shall be conducted on a randomly selected preproduction vehicle by government quality representatives . Examination shall be made at the contractor's or subcontractor's facility during vehicle fabrication and prior to application of primer or paint to evaluate conformance of materials and workmanship to specified requirements. Processing and welding procedures, quality system and inspection records will be evaluated during this examination.

4.4.1.3 First article failure. Failure to pass First article inspection shall be cause for refusal to grant first article approval.(see 6.4.1)

4.4.2 Initial production inspection. The initial production inspection shall be performed on vehicles at random from the first month's production, or the first twenty vehicles produced. The government representative shall make the vehicle selection after all other contractor testing has been performed(see table III and 3.2).

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TABLE III. Inspections and Tests

Title	Requirement Paragraph	Inspection Paragraph	Pre-production 2/	Initial production 1/	Acceptance 2/	Control 2/	CPT 1/
Break-in Run	3.7.3	4.7.1	X	X	X		
5,000 Mile Test	3.2, 3.3, 3.3.1,3.3.2, 3.3.3	4.4.2.1.4, 4.7.2	X	X			
2,000 Mile Test		4.7.3, 4.6.2					X
8 Mile Acceptance Test	3.7	4.7.4	X		X		
50 Mile Control Test	3.7	4.7.5	X	X	X	X	X
Bearing Seals	3.7.1	4.7.6	X	X	X		X
Doors	3.7.2	4.7.7	X	X	X		X
Hatches and Vision Devices	3.7.3	4.7.8	X	X		X	X
Engine	3.7.4	4.7.9	X	X	X		X
Engine Exhaust System	3.7.5	4.7.10	X	X	X		X
Fuel System	3.7.6	4.7.11	X	X	X		X

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TABLE III. Inspections and Tests (Cont.)

Title	Requirement Paragraph	Inspection Paragraph	Pre-production	Initial production	Acceptance	Control	CPT
Fuel System, Grades & Slopes	3.7.6.1	4.7.11.1	X	X	X		X
Fuel Tanks	3.7.6.2	4.7.11.2	X	X		X	X
Hydraulic System	3.7.7	4.7.12	X	X	X		X
Ramp	3.7.7.1	4.7.12.1	X	X	X		X
Transmission	3.7.8	4.7.13	X	X	X		X
Generating System Voltage	3.7.9.1	4.7.14	X	X	X		X
Lights	3.7.9.2	4.7.15	X	X	X		X
Heater	3.7.10	4.7.16	X	X	X		X
Heater Exhaust System	3.7.10.1	4.7.16.1	X	X	X		X
Commun-ications and Electrical Circuits	3.76.11	4.7.17	X	X	X		X
System Voltage Characteristics	3.7.12	4.7.18		X			X

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TABLE III. Inspections and Tests (Cont.)

Title	Requirement Paragraph	Inspection Paragraph	Pre-production	Initial production	Acceptance	Control	CPT
Electro-magnetic Interference Suppression	3.7.13	4.7.19		X			X
STE/ICE Compatibility	3.7.14	4.7.20	X	X	X		X
Controls	3.7.15	4.7.21	X	X	X		X
Adjustment Mechanisms	3.7.16	4.7.22	X	X	X		X
Trim Vane	3.7.17	4.7.23	X	X	X		X
Trim Vane (Control Test)	3.7.17	4.7.23.1	X	X		X	X
Stowed Equipment	3.7.18	4.7.24	X	X		X	X
Spall Liners	3.7.19	4.7.25	X	X	X		X
Engine Cooling	3.8.4.1	4.7.26		X			X
Deaeration and Clearing Capacity	3.8.4.1.1	4.7.26.1		X			X
Drawdown Capacity	3.8.4.1.2	4.7.26.2		X			X

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TABLE III. Inspections and Tests (Cont.)

Title	Requirement Paragraph	Inspection Paragraph	Pre-production	Initial production	Acceptance	Control	CPT
Transmission Lubricant Cooling	3.8.4.2	4.7.27		X			X
Bilge Pumps	3.8.5	4.7.28		X			X
Bilge Pumps (Acceptance Inspection)	3.8.5	4.7.28	X		X		
Speed	3.8.6.1	4.7.29	X	X	X		X
Drift	3.8.6.2	4.7.30	X	X	X		X
Acceleration	3.8.7	4.7.31	X	X	X		X
Engine Starting on Slopes	3.8.8	4.7.32	X	X	X		X
Stopping and Vehicle Control	3.8.9.1	4.7.33		X	X		X
Holding	3.8.9.2	4.7.34	X	X	X		X
Turning	3.8.10	4.7.35	X	X		X	X
Turning (Acceptance Inspection)	3.8.10	4.7.35.			X		
Pivot Steer	3.8.10.1	4.7.36	X	X		X	X

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TABLE III. Inspections and Tests (Cont.)

Title	Requirement Paragraph	Inspection Paragraph	Pre-production	Initial production	Acceptance	Control	CPT
Pivot Steer (Acceptance Inspection)	3.7.10.1	4.7.36.	X		X		
Leakage	3.8.11.1	4.7.37	X	X	X		X
Water Operation	3.8.11.2	4.7.38	X	X		X	X
Trench Crossing	3.8.12	4.7.39		X			X
Vertical Obstacles	3.8.13	4.7.40		X			X
Grade and Slope Operation	3.8.14	4.7.41		X			X
Suspension	3.8.15	4.7.42		X			X
Cupola	3.8.16	4.7.43	X	X		X	X
Cupola (Acceptance Inspection)	3.8.16	4.7.43.	X		X		
Environment-al Conditions	3.9	4.7.44		X			X
Noise Level	3.10	4.7.45		X			X
Toxic Fumes Concentration	3.11	4.7.46	X	X		X	X

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TABLE III. Inspections and Tests (Cont.)

Title	Requirement Paragraph	Inspection Paragraph	Pre-production	Initial production	Acceptance	Control	CPT
Fire Extinguisher	3.12	4.7.47	X	X	X		X
Fire Extinguisher (Control Test)	3.12	4.7.47.1	X	X		X	X
Protective Finish	3.13	4.7.48	X	X	X		X
Marking and Data Plates	3.14	4.7.49	X	X	X		X
Packaging	5.1	4.7.50	X		X		

1/ INSPECTION TO BE PERFORMED BY THE GOVERNMENT AT A GOVERNMENT LABORATORY OR PROVING GROUND
2/ INSPECTION TO BE PERFORMED BY THE CONTRACTOR AT CONTRACTOR'S FACILITY

4.4.2.1 Contractor initial production inspection. The initial production sample vehicle, when completed, shall be inspected by the contractor. Inspection shall include examination as specified in 4.5.1.2. After inspection, the contractor shall submit the vehicle (and all inspection records and certifications) to the responsible Government representative at the contractor's plant for preliminary examination.

4.4.2.1.1 Government preliminary examination. The responsible Government representative shall conduct a preliminary examination of the production sample vehicle as specified in 4.5.1.2..

4.4.2.1.2 Government provisional inspection. Provisional inspection of the production sample vehicle shall be conducted jointly by contractor personnel, representatives of TACOM product assurance directorate and the cognizant Quality Assurance representative. The vehicle

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shall be tested as specified in 4.5.1.3. The contractor shall provide any required assistance and make available his inspection plan, inspection records and certifications pertinent to the vehicle and its components.

4.4.2.1.3 Vehicle disposition. When specified (see 6.2), the production sample vehicle shall remain at the manufacturing facility as a production sample and shall be released at the discretion of the Government. The vehicle shall be serviced and maintained under the vehicle contract during this period. It shall be the last vehicle shipped under contract.

4.4.2.1.4 Government. Government inspection shall include examination as specified in 4.5.1.2 and all Government initial production tests specified in table III. The 5000 mile test shall be performed after successful completion of all other tests at test site specified by the Government.

4.4.2.1.5 Sample size. A minimum of two vehicles shall be subjected to Government inspection.

4.4.2.1.6 Final approval and acceptance. Final approval and acceptance of Government inspection may be withheld until successful completion of Preliminary inspection and acceptance.

4.4.3 Control Test. Control test shall be conducted on one vehicle per month, and shall represent vehicles consecutively produced for that month. The vehicle shall be subjected to the control tests specified in table III. Control test vehicles shall be loaded to full combat weight (27,000 +/- 200 Bs GVW) (see 4.7.5 and 4.7.11.1)

4.4.3.1 Control test frequency. One of the first 10 vehicles produced , and thereafter one per month shall randomly be selected for control testing. The vehicle selected for testing shall be subjected to all control tests specified in table III.

4.4.3.1.1 Vehicle selection. Control test sample shall be randomly selected by authorized Government representative.

4.4.3.1.2 Control test failure. Failure of vehicle to pass any specified control test shall be cause for the Government to refuse acceptance of the production quantity represented until evidence has been provided by the contractor that corrective action has been taken to eliminate the condition that caused the failure.

4.5 Quality conformance inspection..

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4.5.1 Acceptance inspection. After break-in run (see 4.7.1), the vehicle shall be subjected to examinations and tests as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.1 100% inspection. The contractor shall perform acceptance inspection on every vehicle prior to submitting vehicle for Government acceptance.

4.5.1.2 Examination. Visual, dimensional and primary functional examinations of the vehicle shall be made to verify conformance with applicable drawings and this specification.

4.5.2 Acceptance tests. The vehicle shall be subjected to the acceptance tests specified in table III. The vehicle need not be loaded with rated payload.

4.5.2.1 Deficiency sheet. All defects encountered during inspection shall be enumerated on a deficiency sheet for each vehicle. Defects noted on the deficiency sheet shall contain sufficient description to enable the contractor to take corrective action for all defects.

4.5.2.2 Acceptance inspection failure. Failure of a vehicle to pass any specified acceptance test or examination may be cause for the Government to refuse acceptance of the vehicle until evidence has been provided by the contractor that corrective action has been taken.

4.5.3 Government verification of quality conformance. Government inspection to verify quality of vehicles offered for acceptance will be performed at scheduled/unscheduled intervals. The Government shall also verify that a thorough inspection of each vehicle is performed by the contractor and that all defects encountered during inspection are enumerated on deficiency sheets and that all corrective action(s) have been taken.

4.5.3.1 Government inspection. Government shall have a in-plant quality program that meets the policy requirements of Federal Acquisition Regulation (FAR) part 46., Defense FAR Supplement (DFARS) part 246, Quality Assurance (QA).

4.6 Comparison test. The Government may select vehicles at any time during the production period and subject them to comparison test. Comparison test shall be conducted at Government laboratories or proving grounds designated by the contracting officer.

4.6.1 Sample size. Vehicle shall be selected on a spot check basis, in a quantity and at a time specified by the Government. Vehicles selected for comparison test shall not have been previously inspected for conformance to control tests.

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4.6.2 Inspection routine. Sample vehicles will be subjected to examinations as specified in 4.5.1.2 and the comparison tests specified in table III. The 2,000 mile road test (4.7.3.) shall be performed after successful completion of all other tests.

4.6.3 Comparison test failure. Failure of a vehicle to comply with any of the requirements specified herein shall be cause for the Government to discontinue acceptance of vehicles. The contractor shall provide evidence that corrective action has been taken to eliminate the condition which caused the rejection.

4.7 Methods of inspection.

4.7.1 Break-in run. Each vehicle shall be given one of the following break-in runs. Prior to the break-in run, the vehicle shall be serviced, lubricated and operated as specified in 3.8 and 3.8.1.

4.7.1.1 Break-in run (track). The vehicle shall be run on smooth, level, hard-surfaced roads for the distance and speeds specified in table I and over an undulating section of the road as shown in figure 1. The undulating portion of the break-in run shall be accomplished twice in each direction at a speed between 5 and 10 mph during division "a" and at a speed between 11 and 15 mph during division "b". The vehicle shall be operated in reverse after each division as specified in 3.8.3.1. After the break-in run, the vehicle shall be inspected to determine conformance to 3.8.3.3.

4.7.1.2 Break-in run (simulated). In lieu of the break-in run specified in 4.7.1.1, a simulated break-in run shall be accomplished on the Army skid type dynamometer in accordance with table II. The final ten miles of the total break-in run shall be on smooth, level, hard-surfaced roads and over an undulating section of road conforming to figure 1. The undulating portion of the break-in run shall be accomplished twice in each direction at a speed between 5 and 10 mph during division "e" and at a speed between 11 and 15 mph during division "f". The vehicle shall be operated in reverse after each division "a" through "d" as specified in 3.7.3.2. After the break-in run, the vehicle shall be inspected to determine conformance to 3.8.3.3.

4.7.2 5,000 mile test. Sample vehicles shall be loaded with rated payload and road tested as specified in table IV and 4.4.2. Unless otherwise specified, the road test shall be performed by the Government at a Government proving ground. Test shall include verification of reliability, durability and maintainability requirements.

4.7.2.1 Reliability verification. A point estimate value equal to or greater than 850 miles shall be demonstrated during test. The point estimate value will be computed by dividing the total

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cumulative test mileage of all the test vehicles by the total cumulative chargeable failures experienced on these vehicles (see 3.3.1).

4.7.2.2 Durability verification. Durability shall be verified during test at or above a 50% confidence level. The durability requirement will be calculated using a binomial approach requiring a minimum test vehicle sample size of two vehicles (see 3.3.2).

4.7.2.3 Maintainability verification. A maintenance ratio (MR) equal to or less than that specified in 3.3.3 must be demonstrated during test. The MR will be based upon the total cumulative maintenance man-hours with the exceptions of human induced errors and driver/crew checks and services (see 3.3.3).

TABLE IV. 5,000 mile test.

Course	Mileage and Speeds
1/ Hard-surfaced roads	1,000 miles at varying speeds up to maximum
Gravel and dirt roads	2,000 miles at varying speeds up to maximum
Level cross-country	1,000 miles at varying speeds up to maximum
1/ Hilly cross-country	1,000 miles at varying speeds up to maximum
Amphibious operation	One hour minimum at varying speeds up to maximum

1/ NOTE: Hard-surfaced road and hilly cross-country mileage shall include 20 miles each with towed load approximately equal in weight (± 500 pounds) to the test vehicle.

4.7.3 2,000 mile test. Sample vehicle(s) shall be loaded with rated payload and road tested as specified in table V. Unless otherwise specified, the road test will be performed by the Government at a Government proving ground.

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TABLE V. 2,000 mile test.

	Course	Mileage and Speeds
1/	Hard-surfaced roads	200 miles at varying speeds up to maximum
	Gravel and dirt roads	500 miles at varying speeds up to maximum

TABLE V. 2,000 mile test. (Cont.)

	Course	Mileage and Speeds
	Level cross-country	650 miles at varying speeds up to maximum
1/	Hilly cross-country	650 miles at varying speeds up to maximum
	Amphibious operation	One hour minimum at varying speeds up to maximum

1/ NOTE: Hard-surfaced road and hilly cross-country mileage shall include 20 miles each with towed load approximately equal in weight (± 500 pounds) to the test vehicle.

4.7.4 8 mile acceptance test. The vehicle shall be operated for a distance of 8 miles or more over relatively smooth, hard-surfaced roads. Except for equipment and component installation and servicing performed during final processing of vehicle for acceptance, the vehicle shall be completely assembled and serviced with fuel, lubricants and fluids as specified. The equipment and component installation and servicing performed during final processing of vehicle for acceptance will allow omission of such items as floor plates, personnel seat cushions, stowage items, decals, etc. from the vehicle during testing operations. The vehicle need not be loaded with rated payload. During and subsequent to the 8 mile road test, as appropriate, the vehicle shall be subjected to the acceptance tests listed in table III (see 3.7).

4.7.5 50 mile control test. The vehicle shall be loaded to full combat weight and operated for a distance of 50 miles or more over relatively smooth, hard-surfaced roads. During and subsequent to the 50 mile road test, as appropriate, the combat weighted vehicle shall be subjected to acceptance and control tests listed in table III (see 3.7. and 4.4.3.).

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4.7.6 Bearing seals. The final drive and suspension system seals shall be visually examined for evidence of lubricant leaks in excess of a Class I, prior to and after fording and amphibious operations. Lubricants shall be visually examined for water or foreign matter contamination after road testing that follows fording and/or amphibious operations (see 3.7.1).

4.7.7 Doors. The power plant access door and ramp door shall be inspected for completeness of assembly and functional requirements. Doors shall be operated and examined for proper fit and positive latching in opened and closed positions. Door seals shall be examined to verify positive sealing surface to seal contact and seal adhesion (see 3.7.2).

4.7.8 Hatches and vision devices. The cargo hatch, driver's hatch, and commander's hatch shall be inspected for completeness of assembly and functional requirements. The hatches shall be operated and examined for proper fit and positive latching/locking. Installed vision devices shall be examined for proper fit and retention prior to and after vehicle operation. Leakage of water through the vision devices and closed hatches shall not exceed an average of 10 drops per minute, individually, when water sprayed from an open hose with a line pressure of approximately 35 psi and a flow rate of approximately 10 gallons per minute is directed at each hatch and vision device from a distance of approximately three feet for a minimum period of three minutes (see 3.7.3).

4.7.9 Engine. The engine shall be inspected for completeness of assembly and proper installation, and examined for leakage of lubricant, fuel and coolant before and after operation. The engine shall be operated through all speed ranges and exhibit no evidence of low oil pressure, faulty, lubrication, overheating, fuel starvation or leakage of fuel or coolant, or leakage of lubricant at static and dynamic seals in excess of a Class I (see 3.7.4).

4.7.10 Engine exhaust system. The engine exhaust system shall be inspected for completeness of assembly. Tightness of clamping devices shall be examined before and after vehicle road testing. Engine exhaust leakage at connections is acceptable provided clamping devices remain secured (see 3.7.5).

4.7.11 Fuel system. Internal cleanliness of the fuel tanks and lines shall be verified prior to initial fueling of the vehicle. The fuel system shall be inspected for completeness of assembly and proper installation. Connections and sealed items shall be examined for leakage before and after vehicle road testing. No leakage is permitted. (see 3.7.6).

4.7.11.1 Fuel system, grade and slopes. The fuel system shall provide a continuous supply of fuel to the engine and exhibit no fuel leaks while the vehicle is operating on specified grades and slopes. During test, fuel tanks shall be not less than 1/4 full. Control test vehicle fuel tanks shall be filled to capacity (see 3.7.6.1).

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4.7.11.2 Fuel tanks. The fuel tanks shall be filled to capacity at an average rate of 25 gallons per 30 seconds and checked for leakage before and after vehicle road testing (see 3.7.6.2).

4.7.12 Hydraulic systems. Internal cleanliness of the ramp hydraulic system shall be verified prior to initial servicing of the vehicle. The hydraulic system shall be inspected for completeness of assembly and proper installation. Connections and sealed items shall be examined for leakage before and after ramp operation and before and after road testing.(see 3.7.7)

4.7.12.1 Ramp. The ramp shall be inspected for completeness of assembly. The ramp shall be operated and examined for proper fit and positive latching and releasing of the ramp locking mechanism. Ramp rising time from fully lowered to closed position shall not exceed 18 seconds with engine operating at no more than 1200 rpm. NOTE: During the ramp raising test, the transmission range selector shall be in "SL" position and the vehicle brakes locked (see 3.7.7.1).

4.7.13 Transmission. The transmission shall be inspected for completeness of assembly and proper installation, and examined for fluid leakage before and after operation. The transmission shall be operated throughout all speed ranges and exhibit no evidence of overheating, low fluid pressure or fluid leakage at static and dynamic seals in excess of a Class I (see 3.7.8).

4.7.14 Generating system. After the engine has been operated for no less than 20 minutes with the headlights on, the regulator control shall be observed and adjusted if necessary. Voltage at the vehicle batteries shall be 27.3 ± 0.3 with engine operating at 2,150 ± 100 rpm and with a load sufficient to draw 180 ± 5 amperes. When the load is removed, voltage at the vehicle batteries shall be 28.0 ± 0.5 (see 3.7.9.1).

4.7.15 Lights. All internal and external lights shall be inspected for completeness of assembly and operated after vehicle break-in run and road testing to verify proper operation (see 3.7.9.2).

4.7.16 Heater. When heater is not installed, mounting provisions for heater kit installation shall be inspected for conformance to applicable drawings. The heater, when installed, shall be inspected for completeness of assembly and proper installation, and operated to verify functional requirements in MIL-H-62315. Heater controls and heater operation shall be verified in all positions. Care should be taken to assure that heater is allowed to purge itself within 3.5 minutes after operation (see 3.7.10).

4.7.16.1 Heater exhaust system. The heater exhaust system, when installed, shall be inspected for completeness of assembly. Secureness of clamping devices shall be examined before and after vehicle road testing. Heater operation to check for leaks shall be performed after vehicle

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road testing. The exhaust system shall exhibit no leakage during and after heater operation (see 3.7.10.1).

4.7.17 Communication and electrical circuits test. To determine conformance to 3.7.11 electrical circuits supplying current to vehicle equipment shall be tested at all points to assure continuity throughout the circuits. Previously tested circuits shall be functionally tested after installation.. Special kits and equipment, when installed, shall be functionally tested to verify continuity.

4.7.18 System voltage characteristics. System voltage characteristics shall be tested and evaluated in accordance with MIL-STD-1275 (see 3.7.12).

4.7.19 Electromagnetic interference suppression. The vehicle shall be tested for electromagnetic interference suppression in accordance with MIL-STD-462, Notice 3 (see 3.7.13).

4.7.20 STE/ICE compatibility. Vehicle compatibility with specified STE-ICE test equipment shall be physically demonstrated. Function and proper electrical and mechanical interface of STE/ICE with vehicle shall be verified (see 3.7.14).

4.7.21 Controls. All electrical and mechanical controls shall be operated before and after vehicle road testing to verify proper operation (see 3.7.15).

4.7.22 Adjustment mechanisms. All adjustment mechanisms shall be examined before and after vehicle road testing to verify proper operation, maintenance of adjustment settings and that there is adjustment capability within specified ranges. If any adjustment is required, the adjustment setting shall be rechecked to verify that adjustment capability is still within specified range. (see 3.7.16)

4.7.23 Trim vane. The trim vane shall be inspected for completeness of assembly and functional requirements. The trim vane shall be operated and examined for positive latching in the extended and retracted positions (see 3.7.17).

4.7.23.1 Trim vane (control test). The trim vane shall be operated to the extended position and examined for positive latching before and during amphibious operation of the vehicle (table III). After amphibious operation, the trim vane shall be operated to the fully retracted position and examined for positive latching (see 3.7.17).

4.7.24 Stowed equipment. The vehicle shall be completely assembled and equipped with stowage items to verify that all exterior and interior stowed equipment fits in the spaces provided

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and does not interfere with operation of the vehicle. The stowed equipment shall be of the latest configuration available to the contractor. The contractor shall be permitted to open and repackage equipment needed for the test. Unless otherwise specified, the stowed equipment shall be removed from the vehicle after test (see 3.7.18.).

4.7.25 Spall liners. The spall liners shall be inspected for completeness of assembly and proper installation. Spall liners that slide shall be opened and closed to check for binding or interference (see 3.7.19).

4.7.26 Engine cooling. The vehicle shall be serviced as specified in 3.8.1 and operated under the specified conditions of ambient temperature and tractive effort. After stabilization of the cooling system, the engine coolant temperature shall be measured at a point between the coolant outlet from the engine and the transmission oil cooler inlet. Temperature shall not exceed 230°F. The radiator pressure cap shall be pressure tested to verify pressure relief between 12 and 16 psig (see 3.8.4.1).

4.7.26.1 Deaeration and clearing capacity. The engine cooling system shall be tested for deaeration and clearing capacity (see 3.8.4.1.1) as follows:

4.7.26.1.1 Initial test set up. Initial test set up shall be accomplished within the vehicle (See figure 2) to reflect the following:

a. Sight glasses, pressure pick-ups, flow meter and air injection provisions shall be installed in accordance with figure 2.

b. Thermostat shall be installed and blocked open.

c. Pressure cap shall be installed.

d. Coolant shall be fresh water.

4.7.26.1.2 Initial test conditions. Initial test conditions shall be as follows:

a. Engine speed shall be at maximum governed speed.

b. Engine shall be operated at no load.

c. Coolant flow shall be 70 gpm (min.).

d. Coolant temperature shall be 180°F ± 5°F.

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e. Coolant level shall be 12 ± 0.5 gallons.

4.7.26.1.3 Data requirements. Prior to the test, determine the pressure rise across the water pump as a function of coolant flow. The coolant temperature shall be 180°± 5°F with the pressure cap off. Excepting coolant loss, the following data shall be continuously recorded during test:

NOTE: The differential pressure across the pump shall be used to determine the coolant flow (turbine-type meters are not accurate in aerated mediums).

a. Engine speed.

b. Coolant flow.

c. Coolant temperature (radiator inlet).

d. Differential pressure across the water pump (water to pump - 0 to 30 psig) (water from pump - 0 to 60 psig).

e. Surge tank pressure.

f. Air injection rate.

g. Elapsed time.

h. Coolant loss shall be determined after test.

i. Air entrained in coolant from radiator. 1/

j. Air entrained in coolant make-up line. 1/

1/ Monitored visually using sight glass.

4.7.26.1.4 Test procedures. To insure that the cooling system is full, fill it with water and run the engine until the coolant temperature is 190°F. Stop the engine and allow the coolant temperature to fall below 120°F. Add water, as necessary, to the cold fill level (bottom of the fill neck extension). Start the engine and establish the initial operating conditions specified in 4.7.26.1.2. Visually insure that the coolant is free of air. Begin injecting air into the cooling system at 0.15 cfm and 15 psig. Continue for at least 12 minutes to insure stability. Increase the

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air injection rate in 0.15 cfm increments to 0.6 cfm. Inject air at each incremental change for not less than 12 minutes at the 0.6 air injection rate, shut off the air supply. Continue running the engine, until sight glass clears.

4.7.26.1.5 Test failure. The test is a failure, unless all of the following criteria are satisfied:

a. Coolant flow rate shall not fall below 50% of its initial value.

b. Coolant loss shall not exceed four quarts.

c. Cooling system clearing time (elapsed time required for coolant, as observed through a sight glass, to return to its initial clarity after air injection stops) shall not exceed 60 minutes.

4.7.26.2 Draw down capacity. The engine cooling system shall be tested for draw down capacity (see 3.8.4.1.2) as follows:

4.7.26.2.1 Initial test set-up and conditions. Set-up and initial test conditions shall be as specified in 4.7.26.1.1 and 4.7.26.1.2, except the air injection fitting shall be replaced with a hose and valve suitable for draining coolant during the test, and the pressure cap shall be removed.

4.7.26.2.2 Data requirements. Before the test, determine the pressure rise across the water pump as a function of coolant flow. The coolant temperature shall be 180°± 5°F with the pressure cap off. Excepting drained coolant volume, the following data shall be continuously recorded during test:

NOTE: Differential pressure across the pump shall be used to determine coolant flow (a turbine flow meter may not be accurate when the quantity of coolant is reduced).

a. Engine speed

b. Coolant flow.

c. Coolant temperature (radiator inlet).

d. Differential pressure across water pump.

e. Coolant volume drained from system (see 4.15.24.2.3).

f. Air entrained in coolant from radiator. 1/

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g. Air entrained in coolant make-up line. 1/

1/ Monitored visually, using sight glass.

4.7.26.2.3 Test procedure. Insure that the cooling system is full as specified in 4.7.26.1.4. Start the engine and establish the initial operating conditions specified in 4.7.26.2.1. Insure visually that the coolant is free of air. Drain one quart of coolant from the system each minute (minimum) until six quarts have been drained.

4.7.26.2.4 Test failure. The test is a failure, unless the following criteria are satisfied:

a. Coolant flow rate shall not fall below 98% of its initial value.

b. There shall be no indication of entrained air in the coolant.

4.7.27 Transmission lubricant cooling. The vehicle shall be operated under the specified conditions of ambient temperature and tractive effort. After temperature stabilization of the transmission, the transmission lubricant temperature shall be measured at a point in the transmission outlet into the cooler. Temperature shall not exceed 305°F (see 3.8.4.2).

4.7.28 Bilge pumps (acceptance inspection) The bilge pumps shall be operated with sufficient water in the hull to fully submerse the pumps and enable verification of the rate of water removal from the vehicle. Each pump shall discharge water at a rate of not less than 44 gpm with the pumps operating on 28.0 volts maximum (see 3.8.5).

4.7.29 Speed. Prior to testing, the engine shall be warmed up to an operating temperature above 120°F. The vehicle shall then be operated at a speed of no less than 40 mph and exhibit no power plant or suspension malfunction. After testing, the power plant and suspension system shall be examined for damage (see 3.8.6.1).

4.7.30 Drift. The vehicle shall be operated on a dry, level, smooth, hard-surfaced road free of loose material at 25 mph ± 5 mph to check for vehicle drift. Vehicle drift shall not exceed three feet in 100 feet.

NOTE: Steering action to counteract drift is not permitted (see 3.8.6.2).

4.7.31 Acceleration. Prior to testing, the engine shall be warmed up to an operating temperature above 120°F. Then on a dry, level, hard-surfaced road free of loose material with

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transmission in 1-4 range, the vehicle shall be checked for acceleration. The vehicle shall accelerate from 0 to 20 mph in not more than 10.5 seconds (see 3.8.7).

4.7.32 Engine starting on slopes. With the vehicle parked on 30 and 60% slopes in the following positions, the engine shall be stopped and restarted to verify engine starting on slopes:

a. Headed upgrade on 60% slope.

b. Headed downgrade on 60% slope.

c. Left side up on 30% slope.

d. Right side up on 30% slope.

The engine shall restart within one minute after being stopped for not less than one minute after being operated for not less than two minutes under no load at 750 to 1,000 rpm. During test, fuel tanks shall be not less than 1/4 full. Control test vehicle tanks shall be filled to capacity (see 3.8.8).

4.7.33 Stopping and vehicle control. The vehicle shall be driven on a smooth, level, hard-surfaced road free of loose material at a speed of 20 ± 1 mph and stopped by fully applying the service brakes. Stopping distance shall not exceed 40 feet. During stopping test, vehicle drift shall not exceed 3 feet (see 3.8.9.1).

4.7.34 Holding. Ability of the vehicle service and parking brakes to hold the vehicle on a 60% slope shall be verified with the vehicle facing in both uphill and downhill directions. The vehicle shall in each case, remain stationary for one minute minimum with the transmission in neutral and the engine off (see 3.8.9.2).

4.7.35 Turning (Acceptance inspection). With the transmission in "first" range and the vehicle on a smooth, level, hard-surfaced road free of loose material, the vehicle shall be turned a full 360° to the right and then to the left. The diameter described by the outermost edge of the vehicle, while it executes its sharpest possible turn, shall not exceed 45 feet (see 3.8.10).

4.7.36 Pivot steer (Acceptance inspection) With the transmission in "pivot steer" (PV) and the vehicle on a smooth, level, hard-surfaced road free of loose material, the vehicle shall be turned a full 360° to the right and then to the left. The diameter described by the outmost edge of the vehicle, while it executes its sharpest possible turn, shall not exceed 25 feet (see 3.8.10.1).

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4.7.37 Leakage. The vehicle shall be driven into water up to the depth required to float the vehicle. The vehicle shall remain afloat in the water for no less than 15 minutes to check for water leakage into the hull. Leakage rate shall not exceed one quart per minute (see 3.8.11.1).

4.7.38 Amphibious operation. The vehicle, launched amphibiously in calm water of sufficient depth to float the vehicle, shall attain a maximum speed of 3 mph and negotiate a 180° turn within a 30 foot wide corridor. NOTE: Amphibious operation of vehicle shall not be performed until after successful completion of the shallow water crossing leakage test (see 3.8.11.1 and 3.8.11.2).

4.7.39 Trench crossing. The vehicle shall be operated at both a minimum and maximum forward speed over trenches measuring 36 inches deep and 66 inches wide. After trench crossings, the vehicle shall be examined to verify that there is no functional damage (see 3.8.12).

4.7.40 Vertical obstacles. The vehicle shall be slowly operated in forward and reverse gears over vertical obstacles (walls). Height of the obstacles shall be 24 inches for forward gear and 14 inches for reverse gear. After crossing vertical obstacles, the vehicle shall be examined to verify that there is no functional damage (see 3.8.13).

4.7.41 Grade and slope operation. The vehicle shall be operated on specified grades and side slopes and exhibit none of the following during and after test:

a. Malfunction or damage to power train or suspension system components.

b. Overheating of engine or transmission.

c. Faulty lubrication or low oil pressure in transmission and engine.

d. Fuel starvation.

e. Leakage of coolant.

f. Leakage of fuel.

g. Leakage of lubricants at static and dynamic seals in excess of class 1.

Proper oil temperature and pressure shall be verified in each of the four slope positions. Oil pressure readings shall be made with engine operating at 650 to 800 rpm. Vehicle fuel tanks shall be filled to capacity prior to test (see 3.8.14).

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4.7.42 Suspension. The vehicle shall be operated at a speed of 18 to 22 mph and driven over no less than three 6 inch wide by 6 inch high obstacles, spaced 30 feet apart on a flat surface. During impact with the obstacles, there shall be no bottoming out of the track drive sprockets or idler wheels. After testing, the suspension system shall be examined to verify that there is no functional damage to components (see 3.8.15).

4.7.43 Cupola (Acceptance inspection). The commander's Cupola shall be inspected for completeness of assembly and functional requirements. With the vehicle level and the hatch closed, the cupola without gun and ammunition shall be rotated 360°. The cupola shall rotate (after breakaway) without binding under an applied torque of 150 inch pounds maximum. The drag brake and all locking mechanisms shall be checked for proper operation. The azimuth travel lock shall be actuated and the cupola examined before and after vehicle operation to verify that cupola remains secure (see 3.8.16).

4.7.44 Environmental conditions. The vehicle, with and without winterization equipment, shall be operated in specified ambient temperatures and checked for proper performance. The vehicle shall also be stored and exposed to the climatic extremes specified in MIL-STD-810. After exposure to climatic extremes, the vehicles shall be operated and checked for proper performance and any evidence of deterioration that may lead to component failure (see 3.9).

4.7.45 Noise level. Vehicle interior noise level shall be measured while operating the vehicle at 30 mph in 1-4 gear range on a hard-surfaced, straight road with all doors and hatches closed. A sound level meter which meets the requirements of MIL-STD-1474 shall be located in the driver's compartment as specified. Measured sound level shall not exceed 119 decibels (dBA) (see 3.10).

4.7.46 Toxic fumes concentration. Toxic fumes concentration levels shall be measured in the driver and crew compartments per MIL-STD-1472. The vehicle shall be sealed (all hatches and doors closed and the engine access panels installed). The engine shall be operated at 900 to 1200 rpm and the personnel heater, when installed, shall be operated at high heat for a period of not less than 30 minutes. Toxic fumes concentration levels shall not exceed 50 ppm for carbon monoxide or 5 ppm for oxides of nitrogen. Under ambient or test conditions that exceed 55°F, the automatic shut-down of the heater before 30 minutes is acceptable (see 3.11).

4.7.47 Fire extinguisher. The fixed fire extinguisher system shall be inspected for proper charge, completeness of assembly and proper installation (10949802) (see 3.12).

4.7.47.1 Fire extinguisher (control test). The fixed fire extinguisher shall be activated from outside of the vehicle. Maximum effort to discharge system shall not exceed value specified on applicable drawing (10949802) (see 3.12 and table III).

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4.7.48 Protective finish. Protective finish of vehicle and components shall be visually inspected for proper color and coverage in accordance with applicable drawings (see 3.13).

4.7.49 Marking and data plates. Vehicle marking and data plates shall be visually inspected for proper location, legibility and text in accordance with applicable drawings (see 3.14).

4.7.50 Packaging. Inspection of processing for storage and shipment shall be performed on each vehicle in accordance with the applicable specification listed in 5.1 (see 5.1).

5. PACKAGING

5.1 Processing for storage and shipment. Processing of vehicles for storage and shipment shall be in accordance with MIL-C-45360 for model M113A3.

6.0 NOTES

6.1 Intended use. Each vehicle model covered by this specification is capable of crossing inland waterways under its own power and is intended for use as specified below:

6.1.1 M113A3 Carrier. The M113A3 Carrier is used as a personnel carrier.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, revision and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. First article shall be required for initial production (see 3.1 and 4.4).

d. If partially simulated break-in run is authorized (see 3.8.3.2).

e. Level of processing for storage (see 5.1).

f. This Specification does not identify criterion for contact spall (S2 glass or Kevlar).

g. The procuring activity must identify the test parameters in MIL-STD-810 (see 3.9 and 4.7.44)

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6.3 Safety precaution. Caution should be used in handling carbon dioxide extinguisher cylinders. Extreme care should be exercised during reinstallation to avoid tripping fire extinguisher control system since physical injury is highly probable.

6.4 Guidance for the contracting officer on applying first article. The first article sample shall be selected by the Government representative on a preproduction vehicle produced under the production contract for first article inspection. The first article sample should be inspected as specified in 4.4.

6.4.1 First article inspection failure. Deficiencies found during, or as a result of first article inspection, should be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

6.5 Guidance for the contracting officer on applying QCI sampling/acceptance criteria.

6.5.1 QCI. Tests. The preproduction vehicle(s) should meet the quality conformance tests specified in table III.

6.6 Definitions.

6.6.1 Leaks. The following definitions should be used with the classification of defects for leaks:

a. Class 1: Seepage of fluid(as indicated by wetness or discoloration) not great enough to form drop.

b. Class 2: Leakage of fluid great enough to form drops but not enough to cause drops to drip from item being checked /inspected.

c. Class 3: Leakage of fluid great enough to form drops that fall from the item being checked/inspected.

6.6.2. Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.6.3).

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6.6.3 Solid waste. "Solid waste" means (a) any garbage, refuse or sludge from a waste treatment plants, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining and agricultural operations, and from community activities. It dies not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under Section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, Section 23,402).

6.6.4 Stowed equipment. For the purpose of this specification, stowed equipment should be considered as all on vehicle equipment (OVE), basic issue items (BII), and all vehicle components which are not installed at time of shipment.

6.7 International standardization agreement. Certain provisions (see 3.7.11) of this specification are the subject of international standardization agreements, NATO STANAG 2805-A (Edition 3), and ABCA-ARMY-STD-151. When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliatory action through international standardization channels, including standardization offices, if required.

6.8 Recycled materials. The use of recycled materials, which meet the requirements of the applicable material specification without jeopardizing the intended use of the item, is encouraged .

6.9 Mission failure scoring criteria. M113 Family Vehicle Failure Definition and Scoring Criteria may be obtained from Commander, U. S. Army Tank-Automotive Command (TACOM), ATTN: AMSTA-QRA, Warren, Michigan 48397-5000.

6.10 Vehicle rebuild or modification. Vehicles being rebuilt or modified to any of the vehicle model types should meet all requirements covered by this specification.

6.11 Subject term (Keyword) listing.

Amphibious.
Cross country.
Diesel powered.
Personnel Carrier.

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6.12 AMC policy on Acceptable Quality Levels/Lot Tolerance Percent Defectives (AQLs/LTPDs). This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for elimination of AQLs/LTPDs from military specifications.

6.13 Supersession data. This specification supersedes ATPD-2110A dated 30 August 1988.

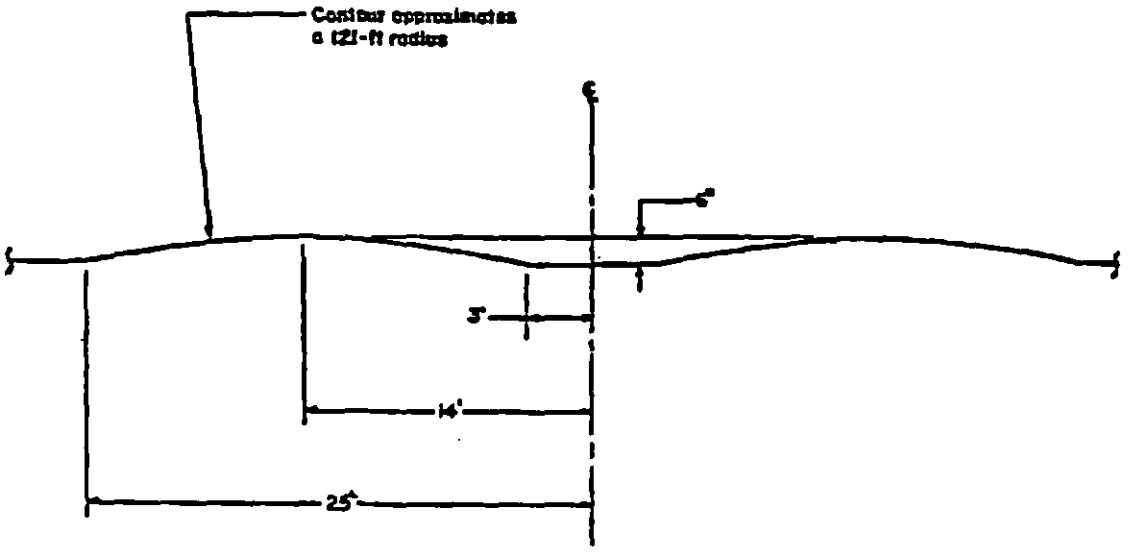
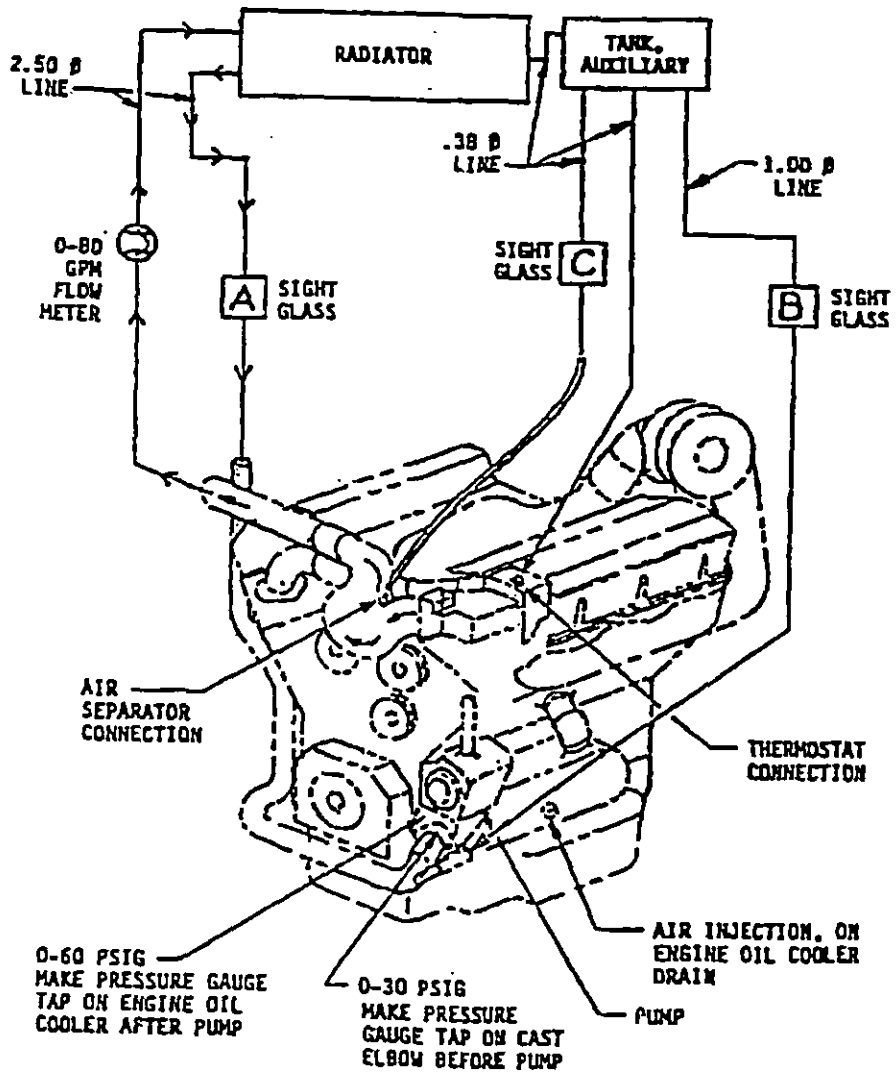


Figure 1. Profile of Test Track (Artificial Bump)

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NOTE: Coolant lines need to be modified to accept gauges, flow meter, and sight glasses.

Figure 2. Coolant System Test Set Up.

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-62746(AT)	2. DOCUMENT DATE (YYMMDD) 930728

3. DOCUMENT TITLE
Carrier, Armored, Full-Tracked: M113A3

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (313) 574-5954 (2) AUTOVON: 786-5954
c. ADDRESS (Include Zip Code) COMMANDER U.S. ARMY TANK-AUTOMOTIVE COMMAND ATTN: AMSTA-GDS WARREN, MICHIGAN 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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